Manufacturing

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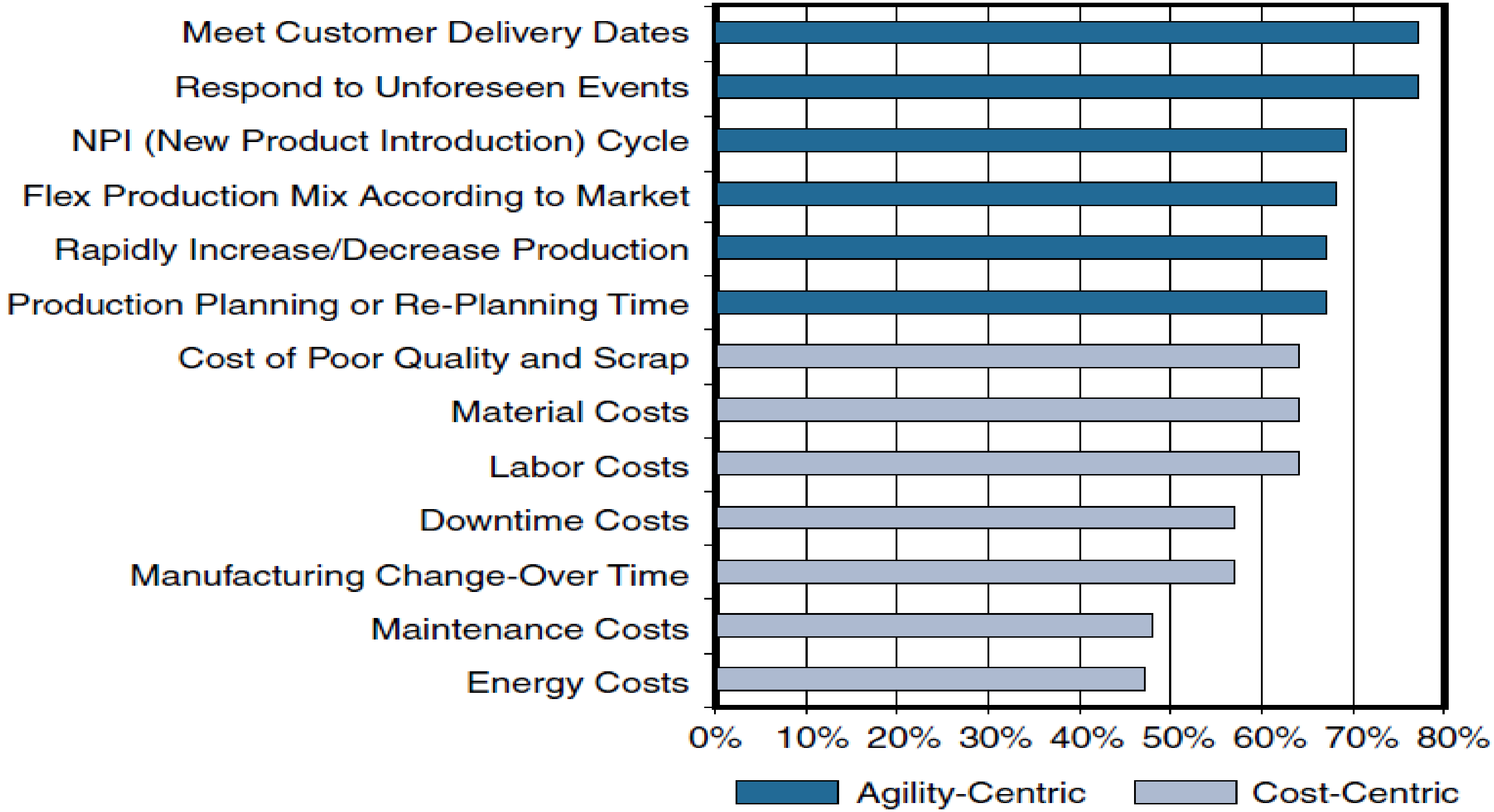
- **An Introduction to Connected Manufacturing:** The chapter opens by examining the technologies that are creating digital disruption in manufacturing. This section also discusses a strategy for the connected factory and the business benefits to manufacturers.
- **An Architecture for the Converged Factory:** Industrial automation and control systems (IACS) networking technologies are converging, with the aid of Ethernet and IP. This section explores a connected factory framework, with a focus on the Converged Plantwide Ethernet (CPwE) architecture that was jointly developed by Cisco and Rockwell Automation.
- **Industrial Automation Control Protocols:** This section discusses the wide variety of networking and control protocols used in manufacturing, including EtherNet/IP, PROFINET, and Modbus/TCP.
- **Connected Factory Security:** This section examines key security considerations in the connected factory and how they can be addressed with the correct design methodology.
- **Edge Computing in the Connected Factory:** The data generated by connected machines is massive. This section examines ways to implement edge computing in the connected factory to improve data management and visibility.

An Introduction to Connected Manufacturing



An Introduction to Connected Manufacturing

- Nearly 75% of US plants are more than 20 years old.
- Their aging assets not only slow innovation but also cost billions in unplanned downtime.
- There are 60 million machines in factories throughout the world. Of them, 90% are not connected, and the vast majority of the machines are more than 15 years old

An Introduction to Connected Manufacturing

- **Data-driven manufacturing:** Big data is changing the face of manufacturing. Manufacturers want access to all data generated from machines to monitor real-time quality control, improve overall equipment effectiveness (OEE), and reduce unplanned downtime. OEE is a well-known metric that indicates manufacturing productivity.
- **OT and IT convergence:** In the context of IoT in a factory setting, operational technology is made up of **programmable logic controllers (PLCs), computers**, and other technology that is often like the technology used in IT but is operated and owned by business operations outside IT. IP networking is enabling **closer integration between machines and factories**, and the line between **factory and enterprise networks** is becoming less distinct. Manufacturers are moving beyond traditional silos and looking for ways to bring their operations together under a single networking infrastructure.

- **Improved technology with lower costs:** New technologies are creating conditions for **scaled, automated, and platform-based machine connectivity, monitoring, and optimization**. In this evolved technology state, machine operations can be viewed as part of a fully connected network system instead of an air-gapped point system. The **convergence of compute, switching, routing, and security** has the potential to drive down the cost of connecting machines.
- **Machine builder OEMs focused on new priorities:** Original equipment manufacturers (OEMs) are facing disruption by new cloud-based providers that intend to provide **Machines as a Service (MaaS),** where machines can be deployed quickly on the plant floor through zero-touch deployment from the cloud, which offers **remote connectivity and monitoring** of those machines. This is driving a new focus on providing better customer experience and emphasizing after-sales products and services. Manufacturers are looking toward near **100% uptime and zero-touch deployments**. They are also exploring ways to control support costs through remote connectivity and monitoring.

An IoT Strategy for Connected Manufacturing

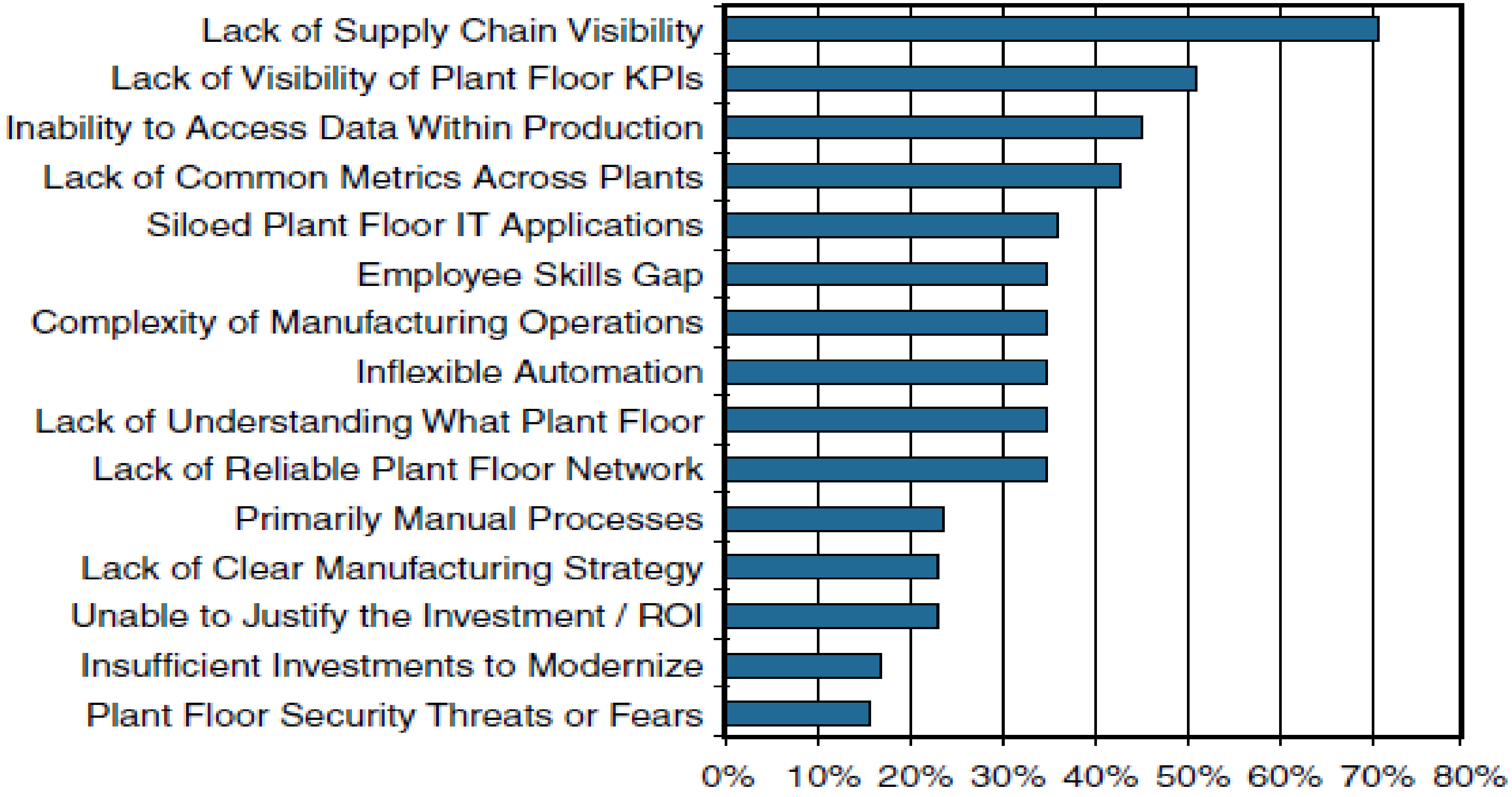


Figure 9-2 *Primary barriers related to information visibility*

Business Improvements Driven Through IoT

- Six Sigma is a set of data-driven manufacturing techniques used to reduce defects aiming for a defect free product 99.99966% of the time.
- In one case where a leading robot manufacturer implemented a real-time data analysis and predictive maintenance application for a leading auto manufacturer, unplanned downtime on several thousand robots was completely eliminated. This saved the manufacturer approximately $40 million in downtime in just a few weeks.

An Architecture for the Connected Factory

- Traditional factory networks were deployed ad hoc. They were isolated and air-gapped from the enterprise IT network
- Network security in the factory was typically limited to an industrial DMZ, leaving the machines mostly unprotected
- Factories rarely deployed network-level security systems that included identity policies and secure remote access tools that allowed plant-level data to be securely extended to the cloud.
- Companies are beginning to tie together their industrial automation and control systems (IACS) with IT applications and analytics tools to provide control and analytics capabilities that are driving operational and business benefits
- Converged Plantwide Ethernet (CPwE) is an architectural framework that provides network services to IACS devices and equipment and promotes secure integration into the enterprise network

Industrial Automation and Control Systems Reference Model

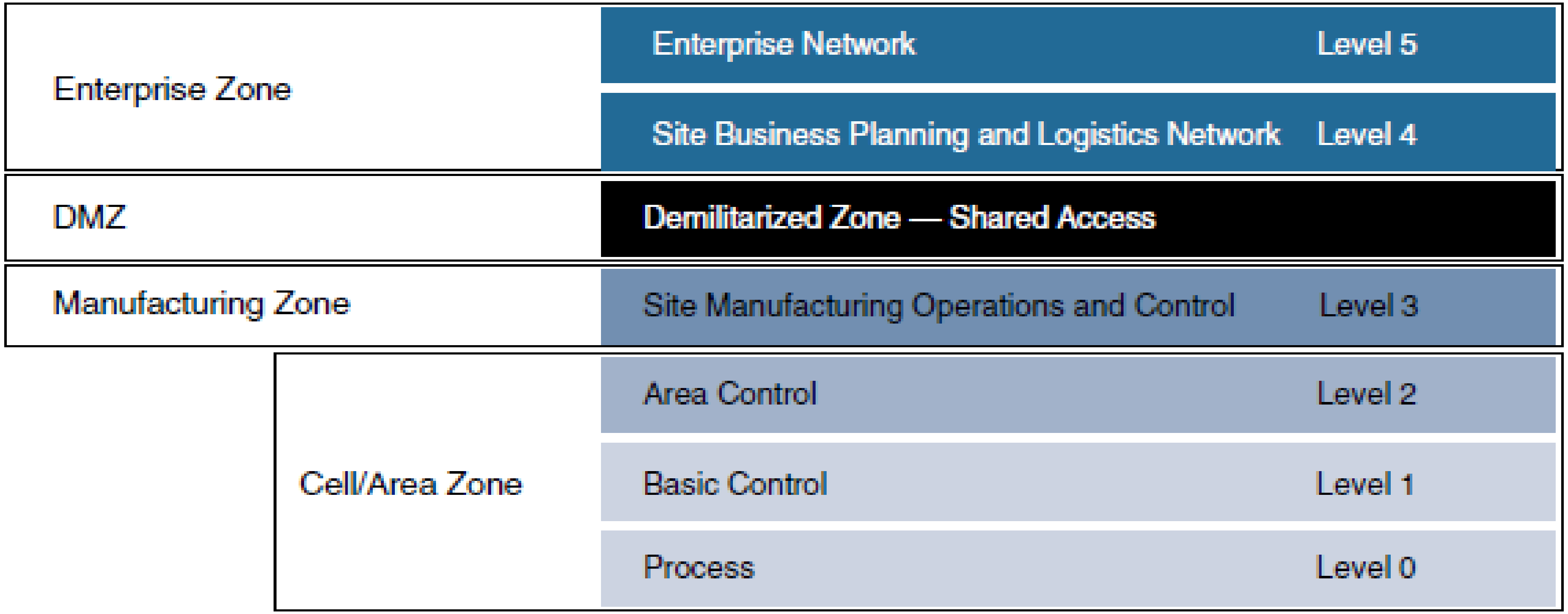


Figure 9-3 *The ISA99 / IEC-62443 IACS Logical Framework, Based on the Purdue Model for Control Hierarchy*

Industrial Automation and Control Systems Reference Model

- **Safety zone:** Systems in the safety zone are typically hard-wired and air-gapped from the IACS network. The safety system's function in this zone is to provide an IACS shutdown (a "stop" button) in case of an emergency. You can think of this as a hardwired fail-safe used to protect personnel and equipment if a dangerous event occurs.
- **Manufacturing zone:** The manufacturing zone is composed of the cell/area zones (Levels 0–2) and site-level manufacturing (Level 3) activities. The manufacturing zone is important because all IACS applications, devices, and controllers critical to monitoring and controlling plant IACS operations are here. To support secure plant operations and functioning of the IACS applications, there is a secure separation of the manufacturing zone and the enterprise zone (Levels 4 and 5).

Industrial Automation and Control Systems Reference Model

- **Cell/area zone:** The cell/area zone is the machine area within a plant. There are typically multiple cell/area zones within a single plant. For example, in an electronics plant, a cell/area may be the assembly process area. The cell/area zone might consist of just a single controller and associated devices, or it could be many controllers on an large assembly line.

This zone has essentially three levels of activity:

- **Level 0: Process:** Level 0 is the "things" level in manufacturing IoT and consists of sensors and actuators involved in the manufacturing process. These IoT devices perform IACS functions, such as moving a manufacturing robot, spraying, driving a motor, and welding. These devices are in communication with the basic control devices in Level 1.

Industrial Automation and Control Systems Reference Model

- **Level 1: Basic control:** Level 1 is where the controllers that direct the manufacturing process live. These controllers interact with Level 0 IoT devices. In discrete manufacturing, a controller is usually a PLC, and in process manufacturing, it is known as a distributed control system (DCS).
- **Level 2: Area supervisory control:** Level 2 includes functions within the cell/ area zone that require runtime supervision and operation. Some examples include HMIs, alarms, and control workstations
- **Level 3: Site level:** The applications and functions at Level 3 include SCADA systems, file servers, control room workstations, scheduling systems, and reporting systems. Note that this level is not a subset of the cell/area zone but is part of the larger manufacturing zone.

Industrial Automation and Control Systems Reference Model

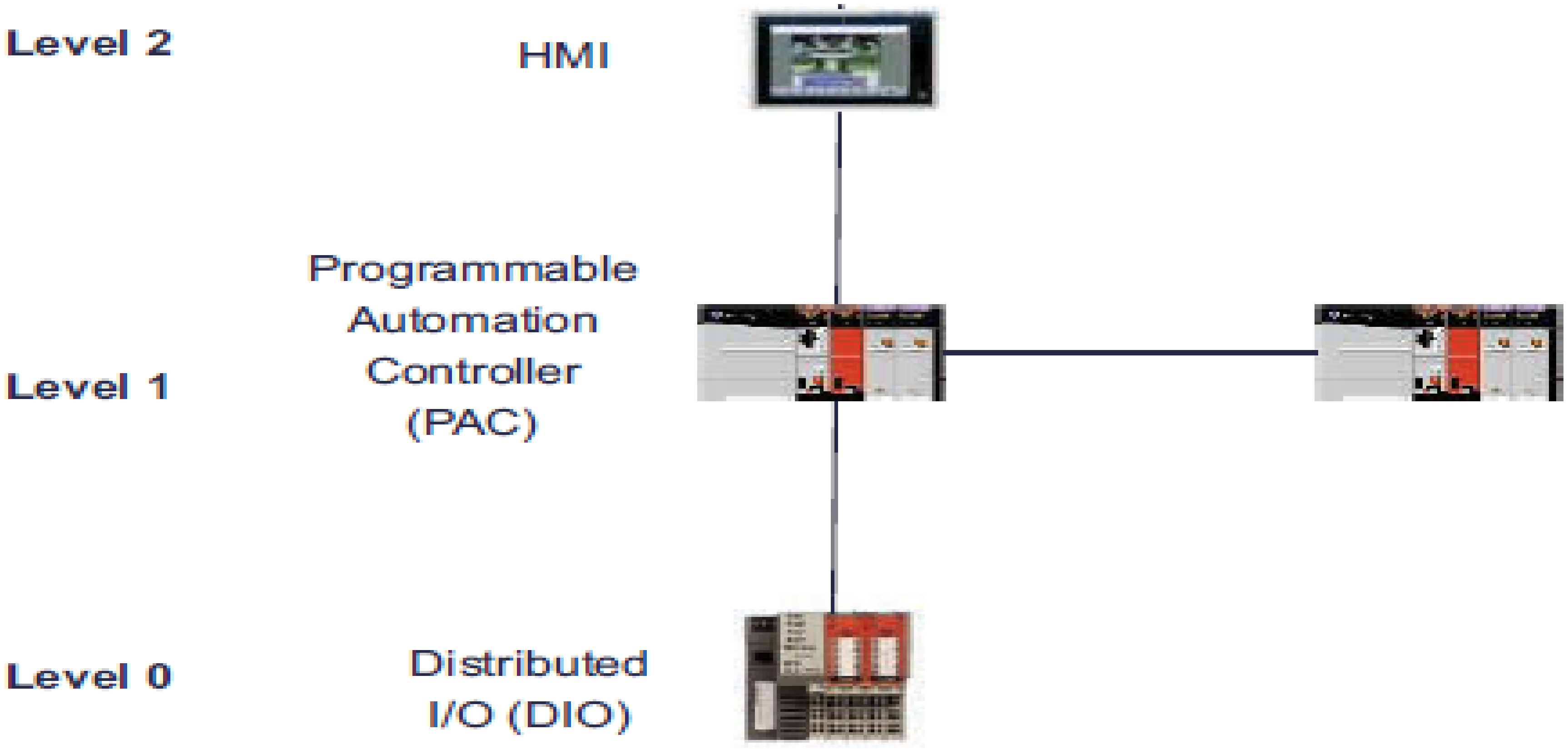


Figure 9-4 *IACS Controller Traffic Flow*

Industrial Automation and Control Systems Reference Model

- **Demilitarized zone (DMZ):** The DMZ is the CPwE demarcation between the plant operational network and the traditional network. DMZ security is critical to plant operations as it protects the machines at the lower level from malicious activity that may occur in the traditional enterprise network.
- **Enterprise zone:** Levels 4 and 5 in the enterprise zone relate to traditional IT/ enterprise networking functions, including file services, Internet connectivity, and email systems.

The CPwE Reference Model

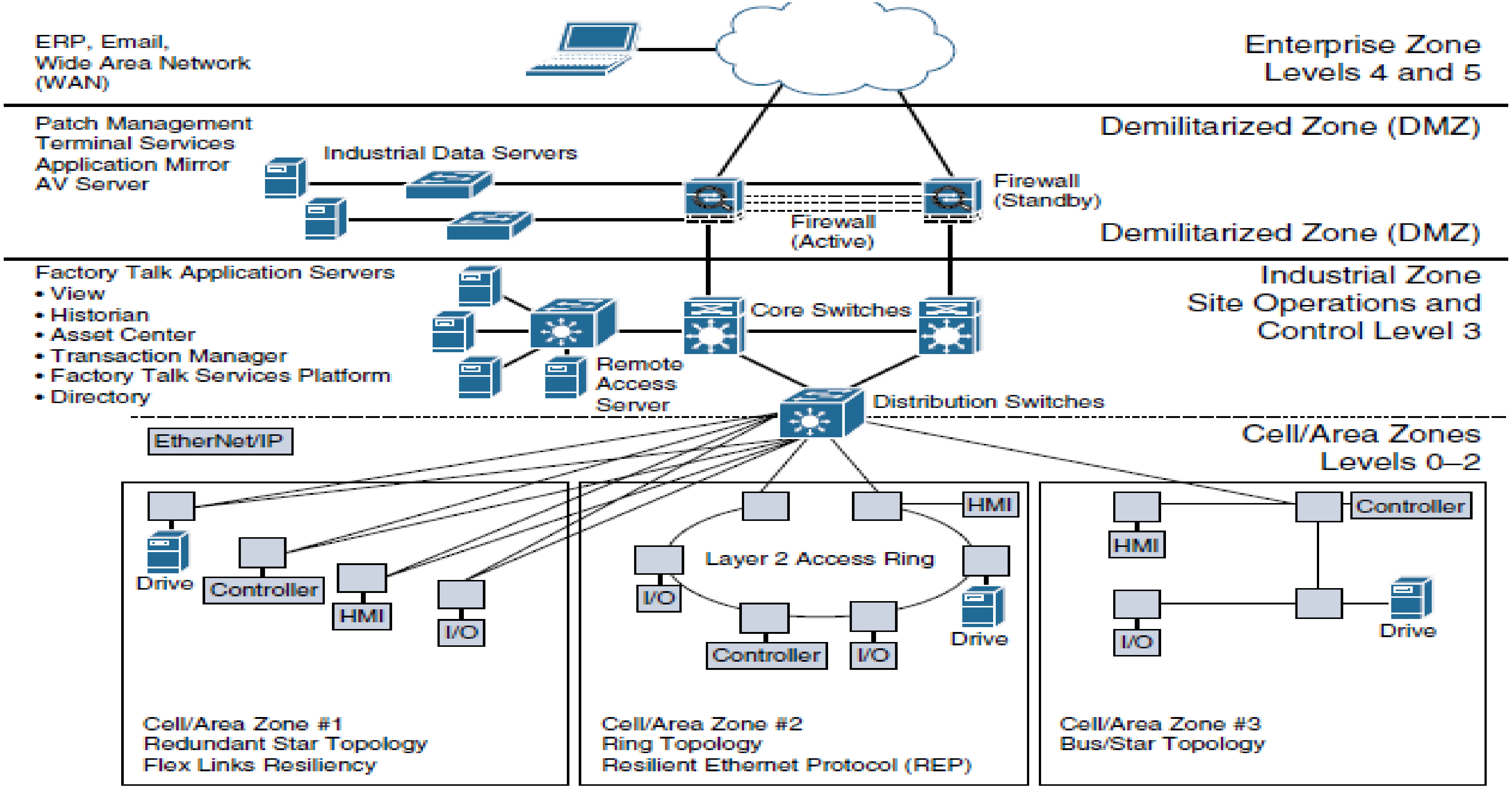


Figure 9-5 *A High-Level View of the CPwE Architecture with Three Different Cell/Area Zone Ethernet Topologies*

CPwE Resilient Network Design

- **Availability:** LAN topology design is critical in supporting IACS application uptime and business continuity. IACS applications have stringent requirements that must be considered for the LAN design, including network availability, performance, and distance between equipment. For critical operations where uptime is crucial, a fully redundant physical path in the IACS Ethernet network topology should be chosen.
- **Predictable performance:** Meeting the predictable, reliable, and real-time traffic requirements of IACS applications is a requirement for successful CPwE deployments.
- **Fast network reconvergence:** In the event of equipment or link failure, network restoration times need to be minimized so that other IACS devices are not impacted by the failure. Typical IACS application interruption tolerance limits are on the order of less than 100 ms, with minimal jitter.
- **Industrial protocol support:** CPwE IACS devices and networking equipment need to support industrial application protocol requirements.

- The following are examples of communication patterns that require network resiliency:

■ Controller to HMI

■ Controller to controller

■ Controller to input/output (I/O; the sensor and controller modules for machines)

■ Controller to variable frequent drives (VFDs; adjustable electromechanical drives to control a motor)

■ Controller to motor control centers (MCCs; used in factories to control a large number of motors from one central controller)

Resilient Ethernet Protocol (REP)

- For each REP segment, one switch is designated as a master node that controls the overall ring. The master node requires three critical pieces of information:
- Identification of the REP control VLAN, which allows the REP control messages to be communicated throughout the segment
- The location of the edges of the REP segment
- The preferred place to break the ring under normal conditions, which is called the “alternate port” (If none is configured, REP automatically selects the alternate port, making the decision nondeterministic.)

Resilient Ethernet Protocol (REP)

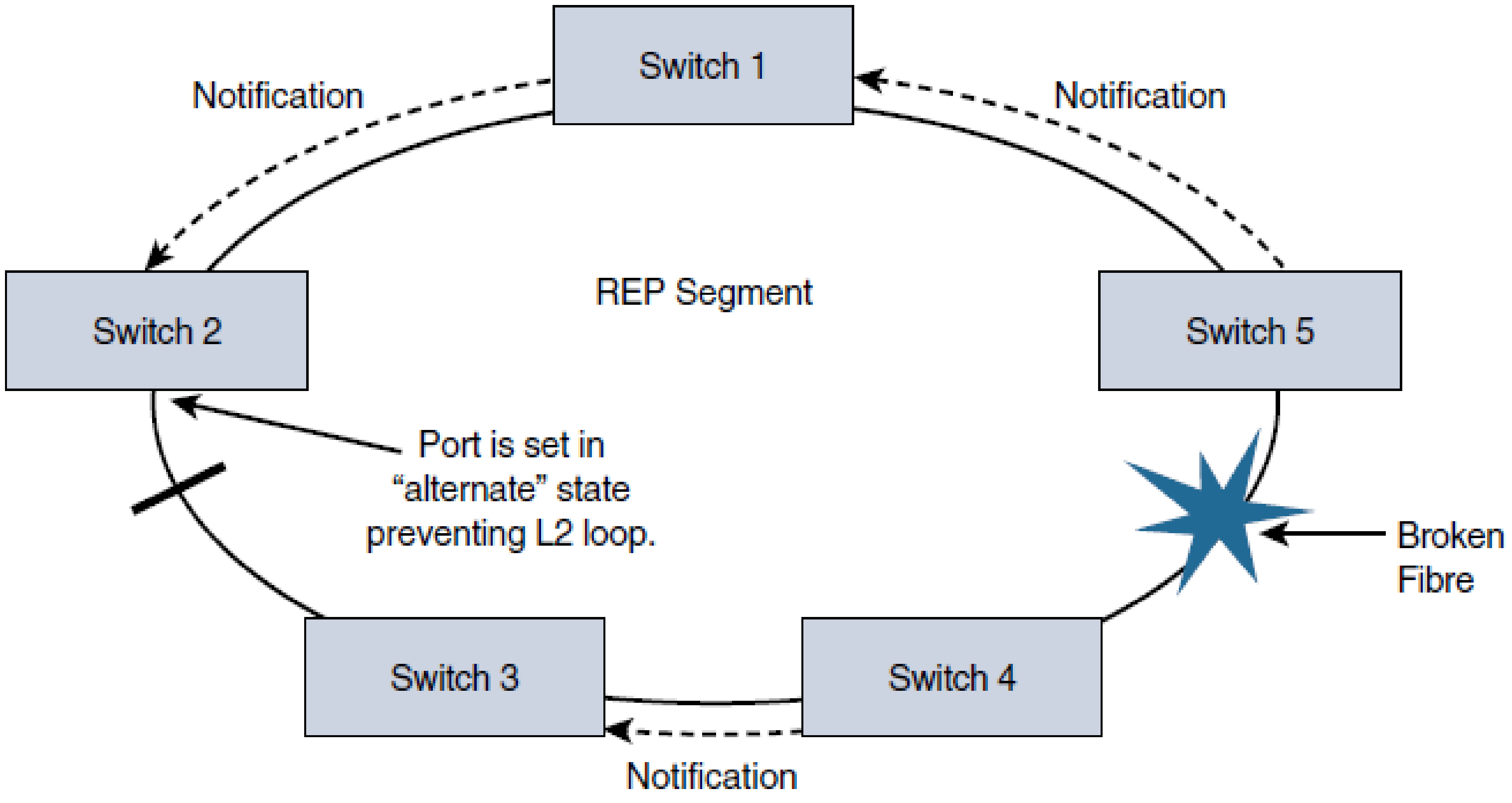


Figure 9-6 *REP Notification When a Topology Change Occurs*

If any other port in the REP segment fails, the alternate port is signaled to change state into the forwarding state, repairing the broken Ethernet ring segment and allowing communications to continue.

REP uses a loss of signal (LOS) detection mechanism to learn of adjacent neighbor failures on the segment. When a switch port detects a REP segment failure (such as a fiber break or a physical switch failure), notification messages indicating a link failure are sent to all the other REP switches.

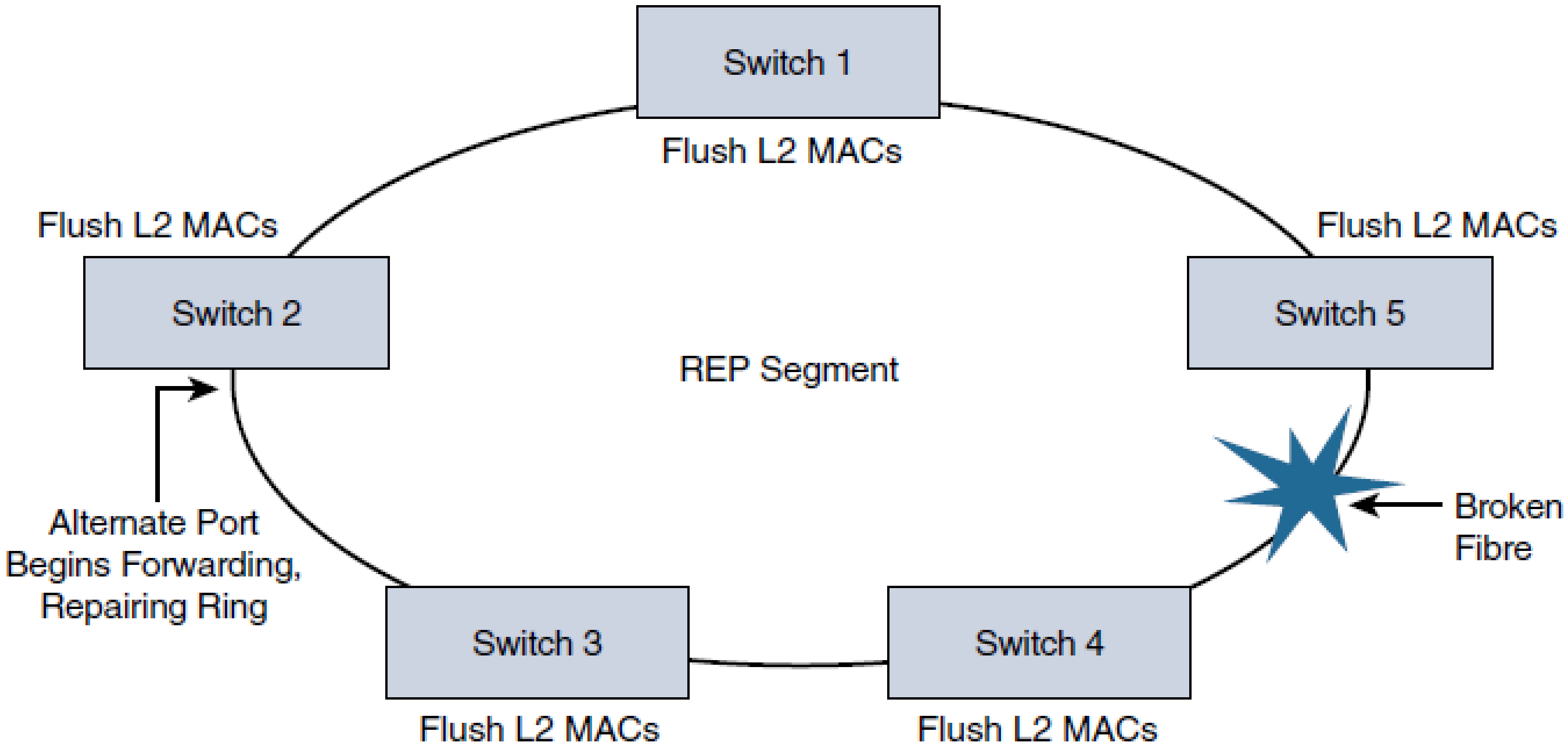


Figure 9-7 *Reconvergence of the REP Ring Segment*

In addition to notifying the alternate port to change to a forwarding state, these notification messages signal that the MAC addresses in content addressable memory (CAM) of all switches must be flushed. In this manner, a new bridging path is formed.

CPwE Wireless Network Architecture

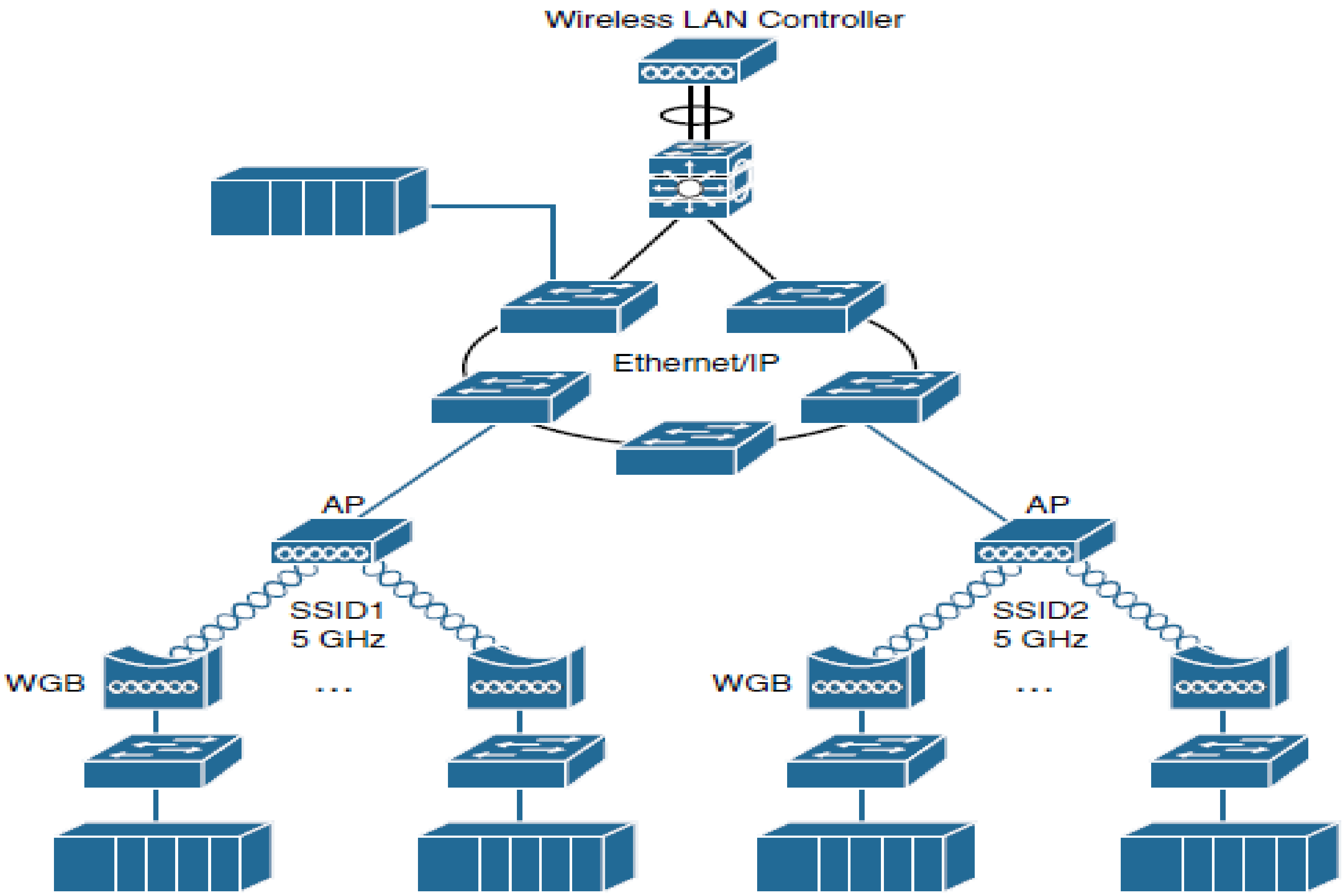


Figure 9-8 *A Factory Wireless LAN Architecture*

CPwE Wireless Network Architecture

- **Fixed-position devices:** Fixed-position devices in the WLAN have a permanent operational location and are also known as “static.” Fixed-position wireless is an alternative to a wired connection for hard-to-reach and remote locations where cabling is too expensive or impossible to install
- **Nomadic devices:** Nomadic equipment stays in place while operating and then moves to a new location in the shutdown state. After relocation, a new wireless connection commonly needs to be established. Examples are process skids, storage tanks, reactors, and portable manufacturing equipment.
- **Operational relocation devices:** Some mobile equipment changes position during an operation, while remaining in the same wireless coverage zone. Examples include rotary platforms and turntables, automated storage and retrieval systems (ASRS), assembly systems, overhead cranes, and similar machinery that uses wireless as a replacement for wired solutions, such as inductive rails and slip rings. These applications may require rapid changes in position and orientation of the wireless client relative to the AP within the coverage area.

Real-Time Location System (RTLS)

- Wi-Fi–based location tracking systems typically include active battery-powered Wi-Fi radio frequency identification (RFID) tags that are attached to machines, skids, vehicles, or other devices that have a measure of mobility within the plant.
- A good example of RTLS in practice comes from one of the world's leading airplane manufacturers. This manufacturer decided to equip all the safety equipment on its planes with RFID tags. When an aircraft goes through maintenance, one job is to inspect each piece of equipment and verify that it is accounted for. Without RFID tags, this job took on average 6.5 hours per plane. With RFID tags in place, the time dropped to 20 minutes per plane.
- The company estimate its gain per year per factory at $100 million.

Industrial Automation Control Protocols

- EtherNet/IP adapts the Common Industrial Protocol (CIP) to standard Ethernet and TCP/IP technology.
- REP is used as a resiliency mechanism between the industrial Ethernet switches (IESs) to pass CIP Class 1 (real-time Ethernet) and Class 3 (TCP) messages
- EtherNet/IP also specifies a redundancy protocol known as Device Level Ring (DLR), which is used when the system requires continuous operation and is able to achieve high-speed reconvergence in the case of a ring break.

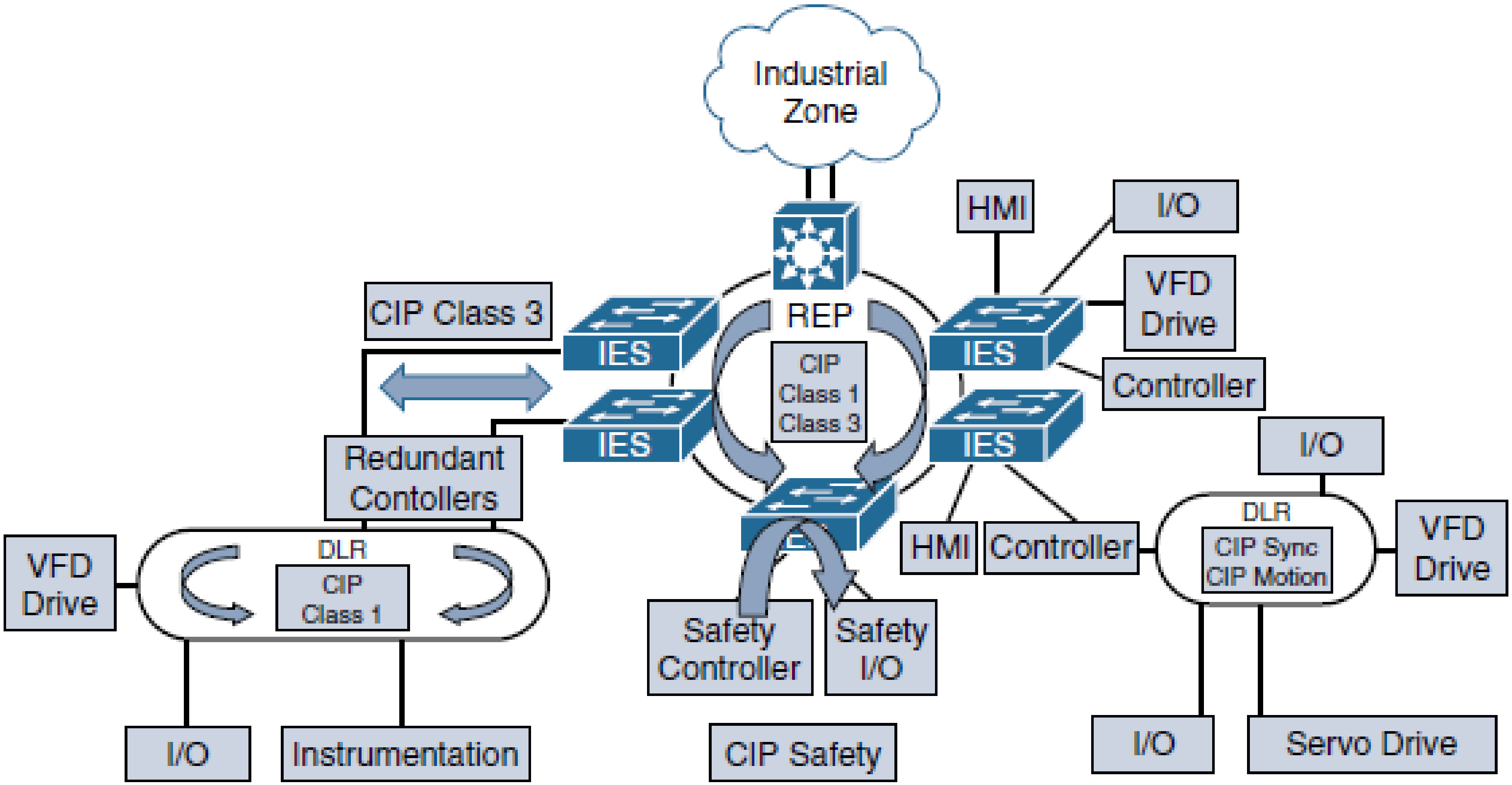


Figure 9-9 A Factory Network Based on EtherNet/IP

PROFINET

- PROFINET (Process Field Net) is a widely used industrial technology for the exchange of data between controllers and devices.
- exchanges messages in a deterministic manner over high-speed Ethernet links
- PROFINET is able to send and receive data directly to the application layer, without having to wait for processing in the TCP/IP stack
- PROFINET is fully compatible with standard IEEE 802.3 Ethernet, which means regular Ethernet devices can coexist with PROFINET I/O devices and controllers on the same segment
- PROFINET also has some significant differences from standard Ethernet. For example, PROFINET is a deterministic protocol, which means frames are sent and received at specific times. This is especially important in discrete manufacturing, when a controller needs to send a message to a device to stop or change operation.
- PROFINET networks are designed to support real-time PROFINET communications with minimal latency, while supporting network resiliency at the manufacturing plant floor.

PROFINET

- PROFINET architectures consist of the following:
  - **Industrial automation devices**: These include robots, sensors, actuators, and drives.
  - **HMIs:** HMIs provide visual status reports and control of the industrial automation devices.
  - **Controllers:** Examples include PLCs and distributed I/O devices.

A well-designed PROFINET architecture provides the following operational benefits:

- It **reduces the risk of production downtime** through the use of a resilient network architecture capable of network convergence
- It **improves plant uptime** through validated reference architectures, with a focus on application availability
- It **enriches critical information access** from machines and applications through better managed network resources.

Media Redundancy Protocol (MRP)

- Determinism and network performance are key requirements for PROFINET stability in the cell/area zone. Determinism in industrial automation ensures that Ethernet frames are sent and arrive when required
- PROFINET device is responsible for scheduling and transmitting the Ethernet frame, the network's main impact on a system's determinism is based on the following performance characteristics:
  - **Latency:** The average amount of time a message takes to be transmitted and processed from originating node to destination node
  - **Jitter:** The amount of variance in the latency
  - **Packet Loss:** The number of packets, usually expressed as a percentage, lost in a transmission from one device to another
- Industrial automation networks need to have low levels of latency and jitter, and they need to support reliable data transmission for real-time applications

- Industrial automation networks must adhere to the following requirements for real-time applications:
- **Machine and process cycle times:** This includes the frequency with which the industrial automation application moves from one operation to the next.
- **Request packet interval (RPI) or I/O update time:** This is the frequency at which input and outputs are sent and received.
- **Packet-loss tolerance:** This is the number of consecutive packet intervals lost before an application generates errors or fails into a safe state.

Media Redundancy Protocol (MRP)

- PROFINET supports a ring resiliency protocol known as Media Redundancy Protocol (MRP).
- MRP is an industry protocol defined in the IEC 62439-2 standard
- MRP allows rings of industrial Ethernet switches to overcome a single segment failure with recovery times similar to those of REP
- MRP is suitable for industrial Ethernet applications and is natively supported in PROFINET.
- PROFINET-compliant industrial Ethernet switches support two roles in an MRP ring.

Media Redundancy Protocol (MRP)

- In a ring topology, only one switch or industrial automation device can act as a media redundancy manager (MRM), and all other devices act as media redundancy clients (MRCs)
- The purpose of the MRM is to keep the ring loop free and provide redundancy when failure happens.
- The MRM does this by sending control packets or test frames from one ring port and receiving them on its other ring port.
- During normal operation, the control packets are received, and the MRM keeps a port blocked to prevent a loop
- If the MRM does not receive its own control packet, this means the loop is not intact, and a network failure has occurred. The MRM informs the MRCs about the network failure
- It unblocks its port and starts forwarding to provide connectivity until the network failure is resolved.

The PROFINET Architecture

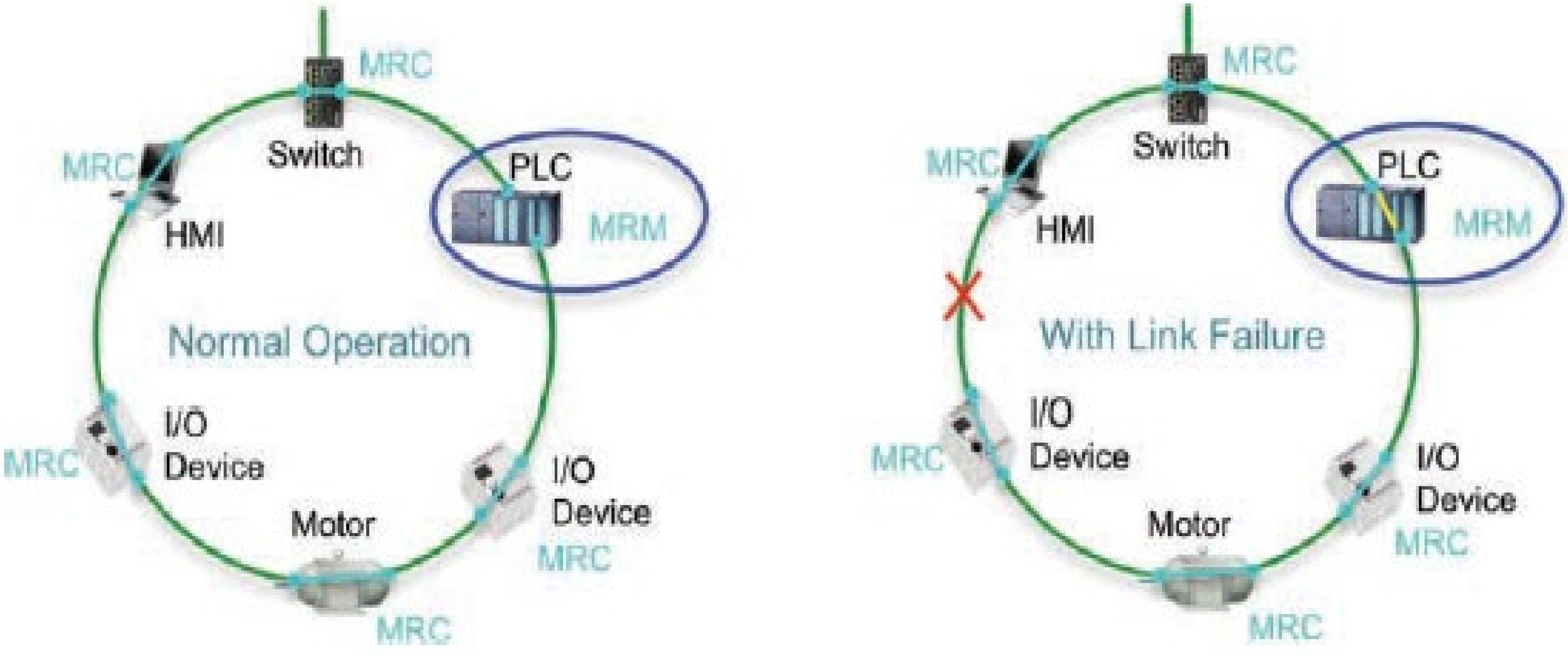


Figure 9-10 *PROFINET MRP Operation*

Table 9-1 *A Comparison of Ethernet Ring Resiliency Protocols*

Protocol	Topology	Number of Nodes	Typical Convergence	Comments
802.1D STP	Any	Max 7 hops	50 s	Not suited for industrial automation due to slow convergence time, which affects real-time applications
Rapid Spanning Tree (802.1w)	Any	Max 20 hops	~2-3 seconds –6 s	Not well suited for ring topologies
MRP	Ring	50	30–500 ms	Part of PROFINET
ITU G.8032	Ring	16 recommended (250 max)	50 ms	ITU standard, similar to REP
DLR (Device Level Ring)	Ring	50	3 ms	Predictable convergence
REP	Ring	Unlimited	50–250 ms	Cisco proprietary but widely deployed in manufacturing, utilities, and other industrial use cases

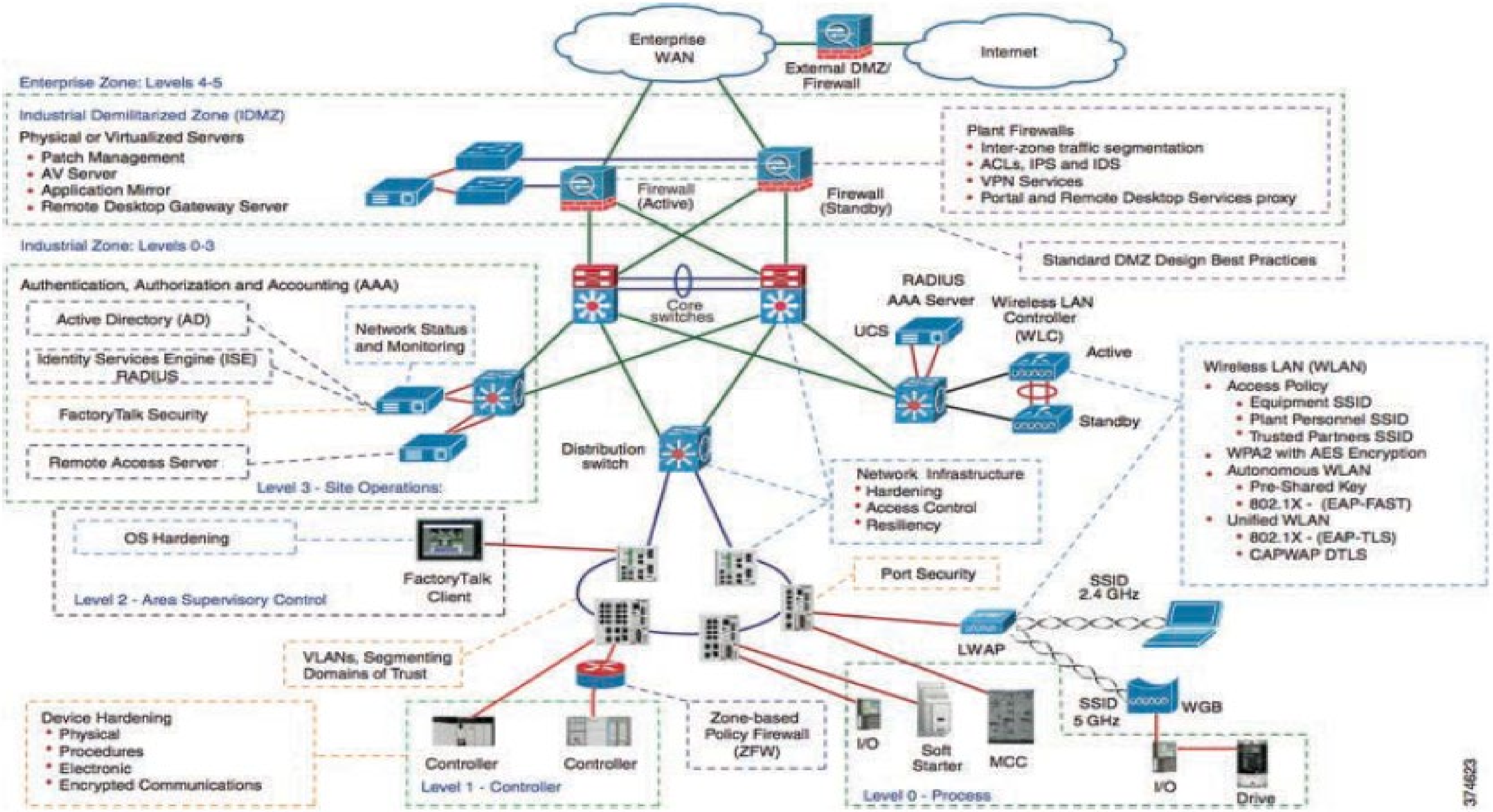


Figure 9-11 *Industrial Network Security Framework*